

ENERGY CHANGES – HYDROGEN FUEL CELLS

KEY VOCABULARY

Fuel cell: uses fuel + oxygen to produce a voltage continuously.

Hydrogen fuel cell: reacts hydrogen with oxygen to make water.

Oxidised: hydrogen loses electrons at the electrode.

Product: only water — no carbon dioxide.

COMMON MISCONCEPTIONS

~~Fuel cells need recharging.~~

✓ They run as long as fuel is supplied.

~~Fuel cells release CO₂.~~

✓ A hydrogen fuel cell only makes water.

~~Hydrogen is easy to store.~~

✓ Hydrogen is hard and costly to store safely.

OVERALL REACTION



Hydrogen is oxidised at the negative electrode.

Oxygen is reduced at the positive electrode.

The only product is water.

Chemical energy → electrical energy.

1

OVERALL EQUATION

Write the overall equation for a **hydrogen fuel cell**. [1 mark]

2

THE PRODUCT

State the only product made in a hydrogen fuel cell. [1 mark]

3

ADVANTAGES

Explain two advantages of fuel cells over rechargeable batteries. [2 marks]

4

DISADVANTAGES

Suggest two disadvantages of using hydrogen fuel cells. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

In the cell, hydrogen is ____ and the only product is ____.

oxidised

reduced

water

carbon dioxide

6

EVALUATE

Evaluate whether fuel-cell cars are truly 'clean'. [2 marks]